

I've reviewed your detailed documentation and the screenshot. You've done an extraordinary amount of systematic debugging here—this is genuinely impressive work.

After analyzing your CSS attempts and the libadwaita default CSS you extracted, I believe I see what's happening.

## The Core Problem

**The button itself is square (24×24). The *visual* circle is an oval because of how Nautilus sizes the > image child relative to the button.**

Here's the key insight from your default libadwaita CSS:

```
windowcontrols > button > image {  
  background-color: color-mix(...);  
  border-radius: 100%;  
  padding: 4px;  
  /* NO min-width, NO min-height */  
}
```

The image's size comes from:

1. The icon's intrinsic size (16×16 for symbolic PNGs)
2. Plus padding: 4px on all sides → 24×24 **if** the icon is perfectly centered

**But Nautilus appears to be giving the image more horizontal space than vertical space** inside the button. This could happen if:

### Most Likely Culprit: -gtk-icon-size not being honored for PNGs

In GTK4/libadwaita, -gtk-icon-size works reliably for **SVG** icons but has inconsistent behavior with **PNG** symbolic assets. Nautilus may be using a different icon loading path that bypasses your -gtk-icon-size: 14px.

## The Solution That Should Work

Based on your research (especially Discourse thread #23354 where they *enlarged* buttons successfully), the missing piece is **forcing the image to be exactly the same size as the button's content area**:

```
/* Reset everything on the button */
windowcontrols button {
  padding: 0 !important;
  margin: 4px !important;
  min-width: 24px !important;
  min-height: 24px !important;
  background: none !important;
  border: none !important;
  box-shadow: none !important;
}

/* Make the image fill the entire button */
windowcontrols button > image {
  -gtk-icon-size: 24px !important; /* Match button size, not smaller */
  padding: 0 !important;
  margin: 0 !important;
  min-width: 24px !important;
  min-height: 24px !important;

  /* Your styling moves to the image */
  background-color: rgba(15, 21, 37, 0.5) !important;
  border-radius: 100% !important;
  border: 1px solid rgba(0, 153, 255, 0.12) !important;
}

/* Hover states on the image */
windowcontrols button:hover > image {
  background-color: rgba(0, 153, 255, 0.22) !important;
  border-color: rgba(0, 153, 255, 0.45) !important;
}

windowcontrols button.close > image {
```

```
background-color: rgba(239, 68, 68, 0.22) !important;
border-color: rgba(239, 68, 68, 0.4) !important;
}
```

```
windowcontrols button.close:hover > image {
  background-color: rgba(239, 68, 68, 0.6) !important;
  border-color: rgba(239, 68, 68, 0.85) !important;
}
```

**Why this should work:** Instead of having two separate visual layers (button background + image background competing), you're making the button itself invisible and using the image as the *entire* visual button. The image now has explicit min-width/min-height of 24px with zero padding/margin, so it cannot be an oval.

## If That Still Fails: GTK Inspector Deep Dive

You mentioned enabling the inspector but not using it. This is your fastest path to certainty. Run:

```
GTK_DEBUG=interactive nautilus
```

Then Ctrl+Shift+D to open inspector, click the "Pick" button (arrow cursor), click a window control button, and check:

1. **Size tab** → What are the actual allocated width/height of:
  - a. The GtkButton widget
  - b. The GtkImage child
2. **CSS tab** → For the image, what computed value does -gtk-icon-size show? (If it's crossed out or missing, the property isn't supported for that widget)
3. **Hierarchy** → Does the button have classes like .text-button or .image-button that might be adding horizontal padding?

## Answers to Your Specific Questions

1. **What makes it wider than tall?** The image's intrinsic size + padding is producing a wider-than-tall effective size, likely because Nautilus's icon loader isn't respecting -

gtk-icon-size and the PNG's 16px width forces the button's content area to expand horizontally more than vertically.

2. **Correct specificity?** Your windowcontrols button (specificity 0,0,1,0) should override libadwaita's windowcontrols > button (also 0,0,1,0) if loaded last, but GTK's CSS engine sometimes respects source order differently. Use !important only as a last resort.
3. **GTK-specific sizing properties?** No—you've found all the relevant ones. -gtk-icon-size is the only one that might help here.
4. **Are PNGs resizable via -gtk-icon-size?** Inconsistent. Some GTK4 builds respect it, others don't. This may be the core issue.
5. **Does Nautilus do something different?** Yes—Nautilus uses a custom NautilusHeaderBar subclass (not AdwHeaderBar directly). This may change how window controls are laid out.
6. **all: unset support?** No, GTK4 CSS doesn't support the all shorthand.
7. **aspect-ratio support?** No—that's CSS Grid/Layout Level 3, not supported in GTK4.
8. **GTK Inspector non-interactive?** Not really, but you can run `GTK_DEBUG=interactive nautilus -q` to launch with inspector ready.
9. **Alternative if CSS fails?** Patch Nautilus's UI file (/usr/share/nautilus/ui/nautilus-window.ui) to add `hexand="False"` to the windowcontrols box, or use `GTK_THEME=InterGenOS:highcontrast` as a test.
10. **Why Settings works but Nautilus doesn't?** Settings uses standard AdwHeaderBar; Nautilus uses a custom subclass. The custom subclass may set different layout properties (halign, valign, hexand) on the image child.

## Final Recommendation

Try the solution above (invisible button + image fills button). If that fails, **use GTK Inspector** to get the actual allocated dimensions—that will tell you definitively whether the problem is the button or the image.

Given your "prime directive" of avoiding hacks, if CSS can't solve this, the honest answer is: **this is a Nautilus-specific layout quirk that likely requires patching Nautilus's UI file.** But I'm not convinced we're at that point yet—I think the image-filling-button approach will work.